Online Merchants’ Churn Model: Status

V1.0

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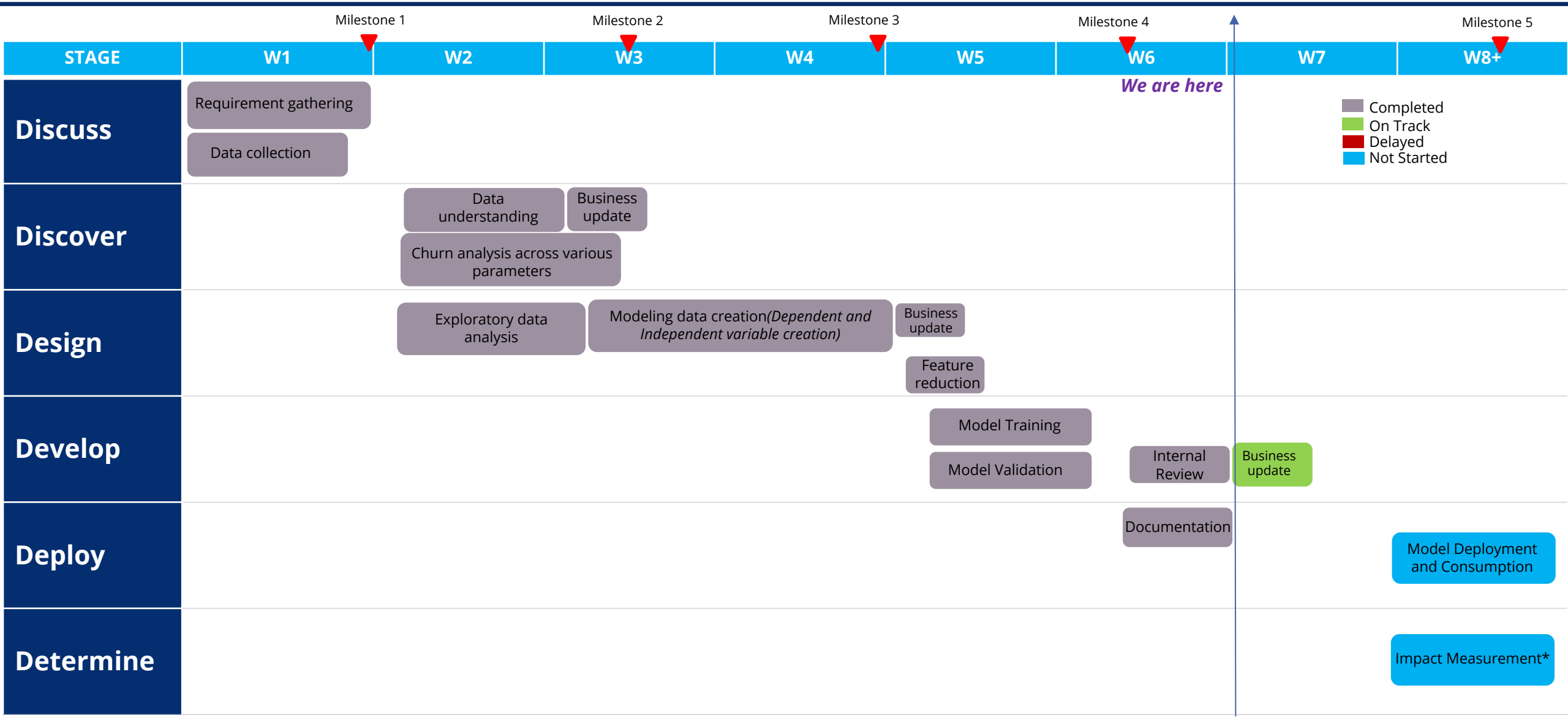
Project Timeline

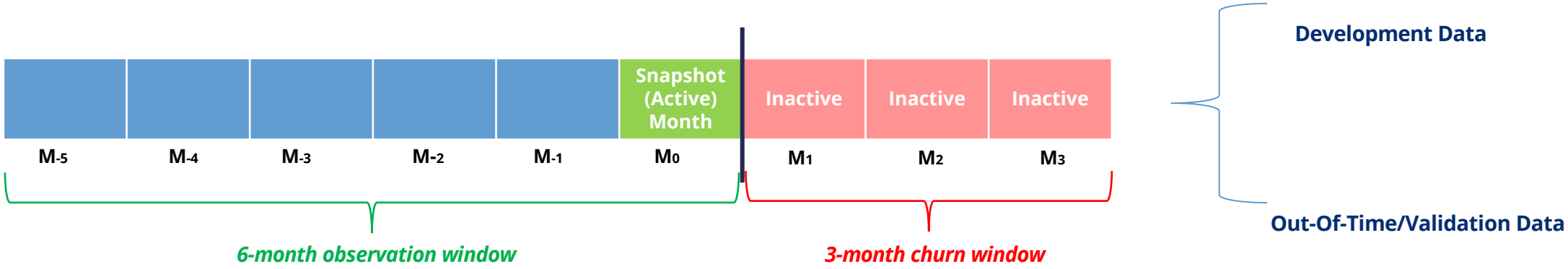


*Impact measurement will be done post receiving feedback from KAMs

Model Design Framework

- Model will be used to predict the **likelihood of currently active merchants to churn** (i.e. inactive for the next 3 months)
- An active merchant as of a snapshot month is observed for its activeness in next 3 months.
- If inactive, then it is labeled as Churned else Non-Churned
- Last 6 months activity will be assessed as of a snapshot month to distinguish Churned from Non-Churned merchants
- Total 10 Snapshot months have been considered to develop the model

Development Framework



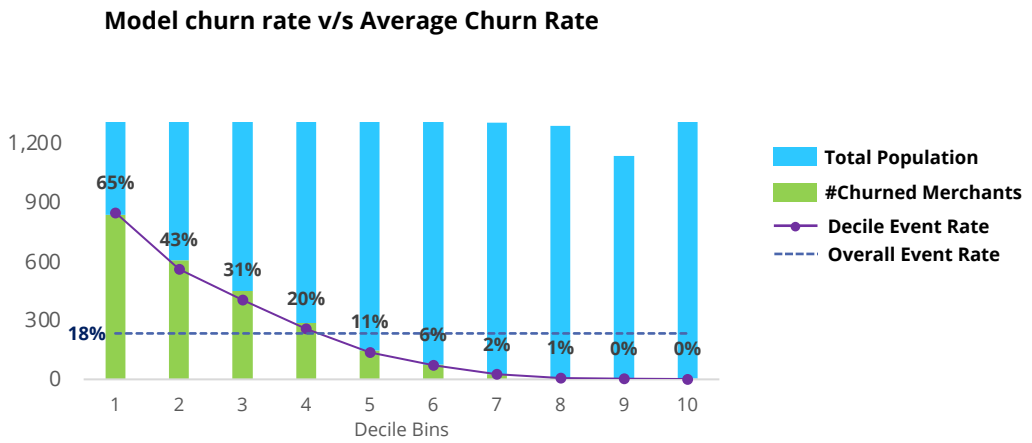
Features would be based on:

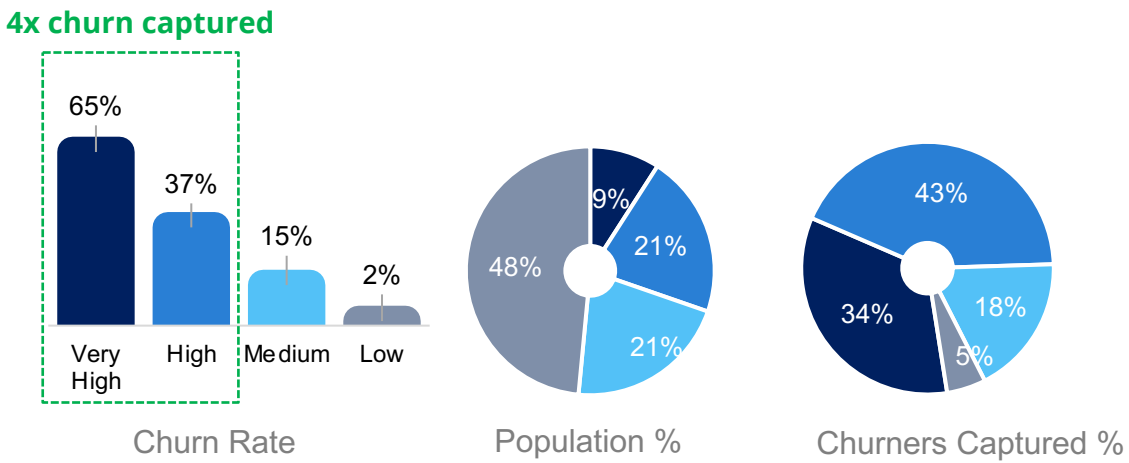
- **Merchant activity**
- **Merchant demogs**
- **Merchant's past complaints and queries**
- **Merchant's activity in Paytm offline space**

Online Merchant Churn Model Performance

Merchant Churn Model

- **30%** population captures **77%** churn
- Top 3 deciles capture **4x churn rate** than average





Top model features

Number of active months in the last 3 months	*Total successful txn in last 6m*	*Average of gap between txns in last 3m*	*Number of days since last transaction activity*	*Average gap between txns with respect to category average gap*
GMV of the latest month	*Monthly limit given to the merchant*	*Daily limit given to the merchant*	*Average of gap between txns in last 6m*	*Tenure of the merchant*

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Next Steps : Implementation and Impact Measurement

Execute, Integrate and Measure Impact

ML Pipeline to be created

- Work with MLOps team to create an automated ML pipeline and execute the model monthly

Integration with Business Team

- Output of the model to be shared with the business team for monthly execution of campaigns
- KAMs to be aligned on the usage of the model output

Campaigns and Impact

- Banners/Campaigns to be focused on Top decile customers
- KAMs to provide feedback on contacted merchants
- Impact measurement – Increase in merchant retention

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